Fiscal Year:	FY 2008	**Task Last Updated:**	FY 10/08/2008
PI Name:	Moore, Steven T. Ph.D.		
Project Title:	Galvanic Vestibular Stimulation (GVS) Augmented Training for Exploration Class Missions		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Sensorimotor Adaptation Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	s.moore@cqu.edu.au	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	212-241-1943
Organization Name:	Mount Sinai School of Medicine		
PI Address 1:	Human Aerospace Laboratory		
PI Address 2:	Department of Neurology		
PI Web Page:			
City:	New York	**State:**	NY
Zip Code:	10029	**Congressional District:**	14
Comments:	NOTE: PI moved to Central Queensland University, Australia, July 2016.		
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	09/01/2005	**End Date:**	04/30/2008
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Team changed to Sensorimotor Adaptation from Smart Medical Systems team in FY2007 per NSBRI (jp 6/08)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NCC 9-58-SM00801		
Performance Goal No.:			
Performance Goal Text:			
	1. Original Project Aims Astronauts returning from spaceflight exhibit deficits in sensorimotor performance that affect posture, gait, gaze and perception of motion; and which present a danger to astronauts during vehicle control, emergency egress, and planetary EVA. We have developed a new system for replicating these sensorimotor effects in normal subjects using Galvanic vestibular stimulation (GVS). GVS uses a small electrical current (<5 mA) passed between electrodes placed on the skin behind each ear to interfere with nerve signals from the balance organs in the inner ear to the brain. The primary aim of the current proposal is to develop training protocols that utilize GVS to enhance crew performance on exploration class missions (GVS-augmented training). 2. Key Findings		

Task Description:	a) We have completed two studies involving 32 normal subjects that demonstrated that acute GVS exposure accurately modeled postural instability and locomotor and gaze dysfunction observed following spaceflight. GVS was well-tolerated, did not induce symptoms of motion sickness, and the effects were reversible (only lasting as long as the current was applied). These results have been published in Experimental Brain Research. b) A pilot study of the subjective response to GVS was conducted at the NSBRI Integration Facility in Houston on May 1st, 2006. Seven veteran astronauts (5 short-term flyers [shuttle], 2 long-term [ISS and Skylab]) participated in the demonstration of GVS technology, which showed that the system generates not only the physiological effects of gravity transitions and re-adaptation to Earth, but also the illusory sensations of motion experienced post-flight. Subjects were exposed to a pseudorandom stimulus, head-driven, and a combination of the two. All shuttle crewmembers (5) stated that the sensations of motion and instability were remarkably similar to that experienced on landing day. The former ISS crewmember felt that the GVS stimulus was not as strong as landing day, and more akin to R+3. These preliminary observations are encouraging as GVS not only accurately reproduced the postural, locomotor and visual post-flight deficits, but also recreated the subjective sensations experienced by astronauts after spaceflight. Moreover, the amplitude of the GVS current used was proportional to the length of exposure to microgravity in the veteran astronauts. Thus, GVS has the potential to provide a high-fidelity ground-based analog of the physiological and perceptual effects of microgravity exposure that can be adjusted to suit mission duration. c) GVS was applied during simulated orbiter landings in both a commercial flight simulator (Airbus A330, Toulouse, France) and the vertical motion (shuttle landing) simulator (VMS) at NASA Ames. The results demonstrated that GVS induced significant roll oscillation of the aircraft during final approach and yaw oscillations during rollout, increased pilot workload, and decreased visual acuity. d) The effects of long-term GVS exposure were assessed in 13 subjects. Repeated (5 sessions) GVS application of 6-min duration during posturography (balance) testing demonstrated that subjects (N=8) did not habituate to the stimulus (i.e., balance performance during GVS remained impaired). However, 5 subjects who were exposed to 1-h of continuous GVS exposure demonstrated habituation during subsequent posturography tests with GVS (i.e., balance performance during GVS returned to baseline [no-GVS] values. 3. Impact of Findings The NASA Human Research Program has identified the development of ground-based analogs of the effects of microgravity exposure on sensorimotor function as a high priority. These studies have demonstrated that the ambulatory Galvanic Vestibular Stimulation (GVS) system shows significant potential as a high-fidelity simulation of postural, locomotor, perceptual and oculomotor deficits observed in astronauts after return from spaceflight. Preliminary studies in commercial and NASA flight simulators demonstrated that GVS induces decrements in pilot performance comparable to that experienced by shuttle pilots during reentry and landing. The results of this project have demonstrated that GVS is an effective, safe, and reversible analog of post-flight sensorimotor dysfunction that could be integrated into astronaut training to improve the fidelity of ground-based mission simulations. Moreover, the experiments on the long-term response to GVS revealed two key findings: 1) repeated short (6 min) applications of GVS did not induce habituation -- this implies that GVS could be used in astronaut training without losing its effectiveness to induce post-flight sensorimotor deficits; and 2) exposing subjects to long-term (60 min) continuous exposure can induce habituation -- this result has interesting future implications for application of GVS to astronaut training, raising the possibility that pre-flight habituation to GVS may provide some protection to the disorienting effects of microgravity on sensorimotor function.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We have developed and validated a model of microgravity-induced sensorimotor deficits using Galvanic vestibular stimulation (GVS). Although conceived as an operational device for astronaut training, the GVS system has potential earth benefits. Our research on the ability of GVS to replicate the postural (balance) problems of astronauts post-landing demonstrated that the GVS system induces a reversible state of imbalance in normal subjects equivalent to patients with vestibular disease. Thus, the GVS system may be used as a model of vestibular pathology to further research in this area. Moreover, patients with vestibular disease often have surgery to remove vestibular input on the affected side, and experience a post-op period of disorientation as the brain adapts to this new sensory arrangement. The potential exists for our GVS system to familiarize patients with the disorienting effects of surgery prior to the actual procedure (analogous to using GVS to familiarize rookie astronauts with the disorienting effects of spaceflight prior to launch). The GVS system is also being used as a model of pilot spatial disorientation in commercial aviation in a collaborative research effort with Qantas Airlines (Australia).
Task Progress:	In the last year of this project we have concentrated on the analysis of data obtained in year 1 and the writing of publications.
Bibliography Type:	Description: (Last Updated: 09/07/2020)
Articles in Peer-reviewed Journals	MacDougall HG, Moore ST, Curthoys IS, Black FO. "Modeling postural instability with Galvanic vestibular stimulation." Exp Brain Res. 2006 Jun;172(2):208-20. PMID: 16432695 , Jun-2006
Articles in Peer-reviewed Journals	MacDougall HG, Moore ST. "Functional assessment of head-eye coordination during vehicle operation." Optom Vis Sci. 2005 Aug;82(8):706-15. PMID: 16127336 , Aug-2005
Articles in Peer-reviewed Journals	MacDougall HG, Moore ST. "Marching to the beat of the same drummer: the spontaneous tempo of human locomotion." J Appl Physiol. 2005 Sep;99(3):1164-73. PMID: 15890757 , Sep-2005
Articles in Peer-reviewed Journals	Moore ST, MacDougall HG, Lesceu X, Speyer JJ, Wuyts F, Clark JB. "Head-eye coordination during simulated orbiter landing." Aviat Space Environ Med. 2008 Sep;79(9):888-98. PMID: 18785358 , Sep-2008

Articles in Peer-reviewed Journals	Moore ST, MacDougall HG, Peters BT, Bloomberg JJ, Curthoys IS, Cohen HS. "Modeling locomotor dysfunction following spaceflight with Galvanic vestibular stimulation." Exp Brain Res. 2006 Oct;174(4):647-59. PMID: 16763834 , Oct-2006